SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Hoo Lee Sheet
Project / Solution Name	Laluan
Name (Leader)	Tan Bing Jie
Email & Phone (Leader)	tanbingjie99@gmail.com, 012-8002983
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Live Prototype URL	Primary: https://hoo-lee-sheet-sdc-2026.replit.app (Replit, synced to main 2f3c4a9) Backup: https://laluan.vercel.app (Vercel, main 2f3c4a9)
Code Repository URL (public / access granted to judges)	https://github.com/Jie-coder/hoo-lee-sheet-sdc2026 (public)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Laluan is a Next.js (React, TypeScript) web app with server-only API routes, deployed on Replit Autoscale (primary) and Vercel (backup, https://laluan.vercel.app). The user uploads a CV or picks a synthetic sample CV. A language model (DeepSeek deepseek-v4.1-flash via the sponsor's KrackedDevs LiteLLM endpoint, else Claude Haiku 4.5 on Replit or Gemini gemini-3.8-flash) parses it into skills and an occupation; KrackedDevs also checks user-entered certificate links; gemini-embedding-001 matches skills to ESCO skills only when exact and synonym matches fail. Deterministic TypeScript code then reads ILO exposure scores, DOSM / JobStreet salary data, WEF scenario parameters, SDG mappings and a curated course catalogue, and produces the ranking, score breakdown, what-if list and pathway, each with its source. No model produces a score, course or link. Changed in Phase 2: an in-app arXiv refresh (POST /api/refresh-signals) stores dated snapshots in Supabase (trend_snapshots), falling back to the last good snapshot, then bundled data; the 29 Oct 2026 snapshot is frozen and bundled for the finals. Be Our Partner (PR #106) is front-end only, with fictional demo partners and prices; it adds no API or dataset and does not affect scoring. Data lives in Supabase (Postgres with pgvector) and in a catalogue built from CSV files.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	**Purpose in Project**	**License**
Next.js 16.4.2; React / React DOM 19.3.0 (Node >=24)	Web app framework and UI library	MIT
Tailwind CSS ^4; shadcn/ui (shadcn ^4.22.0); @base-ui/react ^1.9.0; lucide-react ^1.58.0; framer-motion ^14.4.0; sonner ^2.0.8; next-themes ^0.4.6; tw-animate-css ^1.4.0; class-variance-authority ^0.7.1; qrcode ^1.5.4	Styling, UI components, icons, animation, toasts and PDF ticket QR codes (qrcode, PR #110)	MIT (lucide-react: ISC; class-variance-authority: Apache-2.0)
Vercel AI SDK: ai ^7.0.143, @ai-sdk/anthropic ^4.0.85, @ai-sdk/google ^4.0.99, @ai-sdk/openai-compatible ^3.0.71	Calls to the language and embedding models (Claude Haiku, Gemini; DeepSeek via KrackedDevs LiteLLM through @ai-sdk/openai-compatible)	Apache-2.0
@supabase/supabase-js ^2.118.0; pgvector (Postgres extension on Supabase)	Database client for Supabase, including the trend_snapshots table added in Phase 2 (PR #85); vector search for skill matching	MIT (Supabase JS); PostgreSQL License (pgvector)

Name & Version	Purpose in Project	License
zod ^4.6.7; unpdf ^1.9.1	Schema validation (Zod); PDF text extraction for uploaded CVs (unpdf)	MIT
Google Fonts via next/font/google: Anton, DM Sans, Geist Mono, Noto Sans SC; DM Mono (added in Phase 2, PR #86)	Interface typography; DM Mono for the score-breakdown numbers	SIL Open Font License 1.1
Dev: TypeScript ^5, Vitest ^5.0.5, Testing Library, jsdom ^29.1.1, tsx ^4.23.15, ESLint ^9	Development only: language, tests (758 tests pass on main 2f3c4a9), scripts including npm run kpi (Phase 2, PR #90), linting	Apache-2.0 (TypeScript); MIT (others)
starter-v0 (private repo Jie-coder/sdc-starter, tag starter-v0, commit 57171fe)	Pre-existing team starter code, imported as the first commit (80cc8dc, 19 Oct 2026); declared under Rule 8.2 (see DECLARATION.md, corrected in PR #83)	Team's own pre-existing code (not third-party); repository is private

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini gemini-3.8-flash (Vercel and fallback)	Chat model on Vercel and fallback via getModel(): CV parsing; closest-occupation pick only when exact-name and alias matches fail. Scores, explanations and pathways come from code (no model call)	Free (team leader's own API key)
Google gemini-embedding-001	Skill embeddings: CV skills to ESCO skills when exact and synonym matches fail	Free (team's own API key)
Anthropic Claude Haiku (claude-haiku-4-5) via Replit AI Integrations KrackedDevs LiteLLM endpoint (litellm.rapidscreen.io): DeepSeek deepseek-v4.1-flash	Chat model on Replit via getModel(): CV parsing fallback; closest-occupation pick only when exact-name and alias matches fail KrackedDevs: CV parsing first via getCvParseModel() (PR #100); certificate link check via /api/verify-certificate (PR #103)	Sponsor-provided (billed to organiser-provided Replit credits) KrackedDevs: sponsor-provided (team key and AI API credits)
arXiv API (Cornell University)	AI-trend signal. Changed in Phase 2 (PR #85): called by the app through POST /api/refresh-signals (25 keywords, dated snapshots), no longer only by a manual script. Frozen for the finals (PR #91): last fetch 29 Oct 2026, 16:05:53 (UTC+8)	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel	Backup host: https://laluan.vercel.app, serving main 2f3c4a9. TREND_REFRESH_LOCKED is set in Production for the finals	iad1 (Washington, D.C., USA), default
Replit	Primary host and sponsor platform: Autoscale deployment from the team leader's account (https://hoo-lee-sheet-sdc-2026.replit.app), republished to main 2f3c4a9 for the finals (previous build verified live 31 Oct 2026, 23:03 UTC+8); trend refresh locked (refresh endpoint returns 423); Claude Haiku billed to Replit credits.	Replit Core (sponsor), Autoscale, Asia
Supabase ("Hoo Lee Sheet's Project")	Database + pgvector, server-side only. Phase 2 adds the trend_snapshots table (row-level security on; service role only)	Free tier, ap-southeast-1 (Singapore)
GitHub	Public repo (github.com/Jie-coder/hoo-lee-sheet-sdc2026); pull-request review (no GitHub Actions or CI)	Free
RunPod	Not used (committee confirmed 21 Oct that RunPod is only required for GPU work; Laluan uses hosted model APIs with no self-hosted GPU)	n/a

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Anthropic Claude Haiku 4.5 (claude-haiku-4-5) DeepSeek deepseek-v4.1-flash	Anthropic, via Replit AI Integrations (organiser Replit credits); main model on Replit for occupation matching and CV-parsing fallback DeepSeek, via the KrackedDevs-provided LiteLLM endpoint (sponsor credits); CV parsing first (PR #100) and certificate check (PR #103)	Pre-trained, called via API (no training or fine-tuning by the team)	https://www.anthropic.com/legal/commercial-terms DeepSeek: KrackedDevs sponsor terms for SDC 2026; DeepSeek and LiteLLM terms apply
Google Gemini gemini-3.8-flash	Google, team's own API key; model on the Vercel deployment and fallback	Pre-trained, called via API (no training or fine-tuning by the team)	https://ai.google.dev/gemini-api/terms

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Google gemini-embedding-001 (768 dimensions)	Google; embeddings for skill matching	Pre-trained, called via API (no training or fine-tuning by the team)	https://ai.google.dev/gemini-api/terms
LLMs used in development only: Claude models; DeepSeek; Grok Bot	Claude models via Claude Code and Cursor cloud agents (Phase 1); DeepSeek via OpenCode (Phase 1 and Phase 2); Grok Bot AI agents	Pre-trained, used through the tools in 4.3; not part of the product. The only model added to the product in Phase 2 is DeepSeek deepseek-v4.1-flash (row 1)	n/a (development tools only; see 4.3)

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ESCO v1.2 occupations and skills	https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/frequently-asked-questions	Public	Reuse with acknowledgement under Commission Decision 2011/833/EU	N
ILO Working Paper 140 occupation exposure scores	https://www.ilo.org/sites/default/files/2025-05/WP140_web.pdf	Public	CC BY 4.0	N
Anthropic Economic Index (release 2026-06-26)	https://huggingface.co/datasets/Anthropic/EconomicIndex	Public	Dataset card: "Data released under CC-BY, code released under MIT License" (no CC-BY version stated)	N
WEF Four Futures for Jobs (2026); WEF Future of Jobs Report 2025	https://reports.weforum.org/docs/WEF_Four_Futures_for_Jobs_in_the_New_Economy_AI_and_Talent_in_2030_2025.pdf; https://www.weforum.org/publications/the-future-of-jobs-report-2025/	Public	© World Economic Forum, all rights reserved; used as cited parameters and single figures only	N
arXiv metadata (paper counts for 25 AI-skill keywords)	arXiv API (https://arxiv.org/help/api/index). Phase 2: fetched live by the app; finals snapshot of 29 Oct 2026 stored in Supabase trend_snapshots and data/keywords.csv	Public	CC0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
OpenDOSM open data (wages_occupation)	https://open.dosm.gov.my/data-catalogue/wages_occupation; fetched by scripts/fetch-dosm-demand.ts (PR #30)	Public	CC BY 4.0	N
DOSM salary survey major-group averages	Fallback salary for 8 of 40 roles (PR #50)	Public	Government Open Data Terms of Use 1.0; attribution "Source: Department of Statistics Malaysia"	N
JobStreet Malaysia public salary pages	https://my.jobstreet.com/career-advice; checked by hand into data/salary_by_role.csv (no scraping, no API); salary card only, not scoring; 32 of 40 roles	Public	JobStreet terms: personal, non-commercial use (https://my.jobstreet.com/terms)	N
Course and programme catalogue (198 courses: 184 free, 1 subsidised, 3 HRD Corp-claimable, 10 paid; plus government programmes)	Public provider pages (Coursera, Alison, e-LATiH, Microsoft Learn, Cisco Networking Academy, OpenLearn, HRD Corp, MyFutureJobs and smaller providers). Phase 2 (PR #92): data/course_skill_evidence.csv quotes the syllabus of https://www.coursera.org/learn/pbc 3	Public (links and facts collected by the team)	Each provider's own terms; titles, links, short quotes and own-words summaries only	N
SDG mappings: data/occupation_sdg.csv, data/skill_group_sdg.csv (67 ESCO skill-group rows, Phase 2: PR #87, #92)	Created by the team from the UN SDG targets (https://sdgs.un.org/goals)	Self-collected	Apache-2.0 (part of our repo)	N
Three synthetic sample CVs	Created by the team with AI assistance (PR #3)	Synthetic	Apache-2.0	N (synthetic)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Grok Bot (the team's AI agents)	Phase 1: technical plan, initial repo setup, code review, docs and compliance checks, design (design system, logo, wireframes, synthetic CVs), pitch deck rendering. Phase 2: hand-written changes in PR #83 (DECLARATION.md fix), #85 (live trend pipeline), #89 (favicon) and #91 (freeze), and the hand-edited parts of #84, #86, #87, #88, #90, #92 and #93; review of every OpenCode diff, tests, screenshots and PR text; documentation (README, PRD and report drafts, pitch script and Q&A); pitch deck design in Canva (Canva AI not used: no Canva AI features were used)	Major
OpenCode (DeepSeek model)	Phase 1: light edits from 23 Oct (i18n strings, print receipt draft; parts of PRs #56–#58). Phase 2: most code and tests in #86 (score breakdown, what-if, fit calibration), #87 (pathway SDG total wiring), #88 (about 79% of added lines: uncertainty-band maths and tests), #84 (adoption-slider logic, plumbing and tests), #90 (KPI functions, tests and first draft of the KPI script), the c040 row edit in #92 and the label text change in #93	Phase 1: minor; Phase 2: major
Claude Code	Writing code from T06 onwards in Phase 1, operated via Grok Bot. The Phase 2 PRs (#83–#93) record no Claude Code use	Major (Phase 1)
Cursor cloud agents	Writing code in Phase 1 (T03, T05, scoring engine, T04a results page; PRs #2 onward), operated via Grok Bot	Moderate (Phase 1)
Google Stitch (Google Labs AI UI design tool, https://stitch.withgoogle.com/)	UI / visual inspiration for the website design (no generated code used in the product)	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:
[Maximum 100 words]

Every change went through a GitHub pull request that records which tool wrote each part, with screenshots and test, lint, typecheck and build results. Nothing was merged without Bing Jie's approval (on finals day, Tiffany or Henry approve on GitHub). There is no CI pipeline; main (2f3c4a9) passes 758 Vitest tests, and after each deploy we re-checked the demo numbers on production. In the product, model output is limited to CV parsing, certificate checks and occupation matching from a fixed list; scores, course links and KPIs come from code and cited data.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core (Handbook 9.1.1); finalist credit of US$100 in Replit credits or a Pro plan (Finalist Credits Guide). Finalist credit: yes, used. The finals Replit deployment runs on the finalist Replit credit; see the screenshot in Section 5.	Sponsor platform: Autoscale deployment https://hoo-lee-sheet-sdc-2026.replit.app on the team leader's account; Claude Haiku via Replit AI Integrations billed to these credits. Proof of use: live Replit URL above, plus the Phase 1 proof screenshots in submission/SDC2026_Hoo_Lee_Sheet_Replit_POU.pdf. [Screenshot: Replit Publishing pane] Replit Publishing pane, 31 Oct 2026 23:45 (UTC+8): Live at https://hoo-lee-sheet-sdc-2026.replit.app, republished 23:18 with main 72c1b14.
RunPod	US$25 credit (Handbook 9.1.2; Finalist Credits Guide)	Not used
KrackedDevs	AI API credits, US$20 per teammate (Finalist Credits Guide)	Used: team key on the KrackedDevs LiteLLM endpoint (deepseek-v4.1-flash) for CV parsing (PR #100) and the certificate link check (PR #103)
SDC mentors	Mentoring	Mentor feedback "prove they finished" (docs/programme-proof.md) led to certificate links, the AI-assisted certificate check and the Laluan Certificate of Completion (PR #103)

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is the sponsor platform and pays for Claude Haiku 4.5 through Replit AI Integrations. It is replaceable: the same build runs on Vercel (https://laluan.vercel.app, backup), with Google Gemini on the team's own key. KrackedDevs credits run CV parsing, which falls back to these models, and the certificate check, which shows "pending" without it; the demo link https://demo.laluan.app/example-certificate always verifies, labelled as a demo. The sample CVs are pre-parsed, so the demo works without a live model call, and the frozen 29 Oct trend snapshot is bundled into the app, so the numbers stay the same without Supabase or arXiv.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.
[Maximum 150 words]

Laluan has no login and no accounts. An uploaded CV is sent once to the language model (DeepSeek via KrackedDevs, else Claude Haiku 4.5 or Gemini) to be parsed into skills, and the CV file is not stored. If a user adds a certificate link, the server fetches that public page and sends the URL, course title and up to 8,000 characters of page text to the model; none of it is logged. Users can see and edit every parsed skill before results appear, and a "Delete my data" button clears the session copy. Three pre-parsed synthetic sample CVs mean judges need no real personal data. Supabase (trend_snapshots) holds only public arXiv counts, server-side. API keys are kept in platform secrets, never in the repository or this document. A real launch would still need a full PDPA review.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.
[Maximum 150 words]

All scores and forecasts are estimates and are labelled as such; AI exposure is not job loss. The model can misread a CV, so users can edit every parsed skill. Only 40 ESCO roles are covered, and course and salary data are checked by hand on a stated date. Phase 2 limits: only 398 of 854 skills (46.6%) carry an SDG target; Fit calibration was derived from only three synthetic CVs; the 95% trend range treats correlated keywords as independent, and recent arXiv counts can lag; arXiv allows one request every 3 seconds, so a full re-count takes about 3 minutes (165 s on 29 Oct). Mitigations: last-good and bundled snapshot fallback, a frozen finals snapshot, Gemini as fallback model, pre-parsed demo CVs and Vercel as a second host. We have not run a formal bias test yet.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.

☑ The prototype described here is accessible to judges through the links provided in Section 1

☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[Signature]

[Date] 31 October 2026